

Applications of Change of Variables and First Look at Line and Surface Integrals

Lecture 10 for Multivariable Calculus

目录

Lecture Plan	2
1 Review of Coordinate Systems	3
1.1 Polar Coordinates	3
1.2 Cylindrical Coordinates	4
1.3 Spherical Coordinates	5
1.4 Coordinate Choice Summary	6
2 The General Change-of-Variables Formula	6
2.1 The Formula	6
2.2 Geometric Meaning	7
2.3 Worked Example: Area of an Ellipse	7
3 Large Example: Volume of the n-Ball	8
3.1 Definitions	8
3.2 Low-Dimensional Starting Points	8
3.3 Slicing Recursion	9
3.4 Reducing to a Trigonometric Integral	9
3.5 Computing I_n	10
3.6 Closed Formula with the Gamma Function	11
4 Surface Area of the High-Dimensional Sphere	12
4.1 Shell Argument	12
4.2 Formula	12

5	Average Values and Harmonic Functions	13
5.1	Average over a Region	13
5.2	Harmonic Functions	13
5.3	Example 1: A Linear Harmonic Function	14
5.4	Example 2: A Quadratic Harmonic Function	14
5.5	Same Example by Symmetry	14
6	Physics Applications	15
6.1	Mass and Center of Mass	15
6.2	Torque	16
6.3	Moment of Inertia	17
7	Probability and the High-Dimensional Gaussian	17
7.1	Probability Density	17
7.2	The Standard Gaussian in $\mathbb{R}^n$	18
7.3	Probability Inside a Ball	19
7.4	High-Dimensional Geometry of the Gaussian	19
8	Beginning Line Integrals	20
8.1	Curves and Arc Length	20
8.2	Scalar Line Integrals	20
8.3	Vector Line Integrals	21
8.4	Geometry and Physics of Line Integrals	22
8.5	Green's Theorem	23
8.6	Application: Conservative Vector Fields	27
9	Beginning Surface Integrals	29
9.1	Parametrized Surfaces	29
9.2	Scalar Surface Integrals	29
9.3	Flux Surface Integrals	30
10	Practice Problems	30
11	Summary	32

Lecture Plan

Lecture 10 has two jobs. First, we finish the main applications of change of variables. Second, we open the next topic: integration over curves and surfaces.

1. Review polar, cylindrical, and spherical coordinates.
2. State the general change-of-variables formula and use it cleanly.
3. Work out one large example in detail: the volume of the n -dimensional ball and the surface area of the $(n - 1)$ -sphere.
4. Discuss average values and compute averages of harmonic functions.
5. Explain physics applications: mass, center of mass, torque, and probability. For probability, emphasize the Gaussian distribution in high dimension.
6. Begin line integrals: geometry, physics, Green's theorem, conservative vector fields, and surface integrals.

Guiding Picture

Every integral is still a sum of small contributions:

$$\text{density} \times \text{small size}.$$

The main question is: what is the correct small size?

$$dA, \quad dV, \quad dx_1 \cdots dx_n, \quad ds, \quad dS.$$

Change of variables explains the scaling factors. Line and surface integrals introduce new kinds of small size.

1 Review of Coordinate Systems

1.1 Polar Coordinates

Polar coordinates in the plane are

$$x = r \cos \theta, \quad y = r \sin \theta.$$

The Jacobian matrix is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix},$$

so

$$\det \frac{\partial(x, y)}{\partial(r, \theta)} = r \cos^2 \theta + r \sin^2 \theta = r.$$

Therefore

$$\boxed{dA = r \, dr \, d\theta.}$$

When Polar Coordinates Are Natural

Use polar coordinates when the region or integrand contains

$$x^2 + y^2, \quad \sqrt{x^2 + y^2},$$

or when the region is a disk, annulus, sector, or region bounded by rays.

Example. Compute

$$\iint_{x^2+y^2 \leq a^2} e^{-(x^2+y^2)} \, dA.$$

In polar coordinates,

$$0 \leq r \leq a, \quad 0 \leq \theta \leq 2\pi.$$

Thus

$$\iint_{x^2+y^2 \leq a^2} e^{-(x^2+y^2)} \, dA = \int_0^{2\pi} \int_0^a e^{-r^2} r \, dr \, d\theta.$$

Since

$$\int_0^a e^{-r^2} r \, dr = -\frac{1}{2} e^{-r^2} \Big|_0^a = \frac{1}{2} (1 - e^{-a^2}),$$

the integral is

$$\boxed{\pi(1 - e^{-a^2}).}$$

1.2 Cylindrical Coordinates

Cylindrical coordinates are polar coordinates in the xy -plane together with the height variable:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

The volume element is

$$\boxed{dV = r \, dr \, d\theta \, dz.}$$

The same factor r appears no matter which order of integration is used.

When Cylindrical Coordinates Are Natural

Use cylindrical coordinates when the solid has circular symmetry around the z -axis, or when equations contain

$$x^2 + y^2, \quad r, \quad z.$$

Typical boundaries include cylinders, cones, paraboloids, and disks extruded in the z direction.

Example. Find the volume of the solid

$$x^2 + y^2 \leq z \leq 4.$$

Since $x^2 + y^2 = r^2$, the lower surface is $z = r^2$ and the upper surface is $z = 4$. The two meet when

$$r^2 = 4, \quad r = 2.$$

Thus

$$0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq 2, \quad r^2 \leq z \leq 4.$$

The volume is

$$V = \int_0^{2\pi} \int_0^2 \int_{r^2}^4 r \, dz \, dr \, d\theta.$$

Compute:

$$V = \int_0^{2\pi} \int_0^2 (4 - r^2)r \, dr \, d\theta = 2\pi \left[2r^2 - \frac{r^4}{4} \right]_0^2 = \boxed{8\pi}.$$

1.3 Spherical Coordinates

Spherical coordinates are

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

Here ρ is distance from the origin, θ is the angle in the xy -plane, and ϕ is the angle down from the positive z -axis.

The volume element is

$$\boxed{dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta}.$$

The relation to cylindrical coordinates is

$$r = \rho \sin \phi, \quad z = \rho \cos \phi.$$

When Spherical Coordinates Are Natural

Use spherical coordinates when the solid or integrand contains

$$x^2 + y^2 + z^2 = \rho^2,$$

or when the boundary is a sphere, spherical shell, or cone through the origin.

Example. For the ball $x^2 + y^2 + z^2 \leq R^2$,

$$0 \leq \rho \leq R, \quad 0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$$

Therefore

$$\text{Vol}(B_R^3) = \int_0^{2\pi} \int_0^\pi \int_0^R \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{4}{3}\pi R^3.$$

1.4 Coordinate Choice Summary

Coordinate system	Best signs	Size element
Cartesian	Planes, boxes, graph bounds	$dx \, dy \, dz$
Polar	Disks, annuli, sectors	$r \, dr \, d\theta$
Cylindrical	Cylinders, paraboloids, circular symmetry	$r \, dr \, d\theta \, dz$
Spherical	Balls, spherical shells, cones from origin	$\rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$

Memory Device

The extra factor tells how much a coordinate rectangle or box stretches:

$$r \quad \text{for polar/cylindrical,} \quad \rho^2 \sin \phi \quad \text{for spherical.}$$

Never write only $dr \, d\theta$ or only $d\rho \, d\phi \, d\theta$ for area or volume.

2 The General Change-of-Variables Formula

2.1 The Formula

Let

$$T(u_1, \dots, u_n) = (x_1(u), \dots, x_n(u))$$

be a smooth coordinate change from a region S in u -space to a region D in x -space. The Jacobian matrix is

$$DT = \begin{pmatrix} \frac{\partial x_1}{\partial u_1} & \cdots & \frac{\partial x_1}{\partial u_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial u_1} & \cdots & \frac{\partial x_n}{\partial u_n} \end{pmatrix}.$$

Its determinant measures local n -dimensional volume scaling.

Theorem 2.1 (Change of Variables in $\mathbb{R}^n$). *If $T : S \rightarrow D$ is smooth and one-to-one, with nonzero Jacobian determinant except possibly on boundary pieces, then*

$$\int_D f(x) \, dx_1 \cdots dx_n = \int_S f(T(u)) \, |\det DT(u)| \, du_1 \cdots du_n.$$

One-to-One Caveat

Polar and spherical coordinates are not literally one-to-one on all angle boundaries and at the origin. This is harmless for ordinary integrals because those exceptional pieces have area or volume zero. In practice, choose a standard range and avoid double-counting interiors.

2.2 Geometric Meaning

A tiny coordinate box in u -space has volume

$$\Delta u_1 \cdots \Delta u_n.$$

Near a point u , the map T is approximately linear:

$$T(u + \Delta u) \approx T(u) + DT(u)\Delta u.$$

A linear map with matrix $DT(u)$ changes volume by the factor

$$|\det DT(u)|.$$

Thus the physical small size is approximately

$$|\det DT(u)| \Delta u_1 \cdots \Delta u_n.$$

Core Rule

Change the integrand and change the size element:

$$f(x) \, dx_1 \cdots dx_n \longrightarrow f(T(u)) \, |\det DT(u)| \, du_1 \cdots du_n.$$

2.3 Worked Example: Area of an Ellipse

Compute the area of the ellipse

$$\frac{x^2}{4} + \frac{y^2}{9} \leq 1.$$

Use the change of variables

$$x = 2u, \quad y = 3v.$$

Then

$$\frac{x^2}{4} + \frac{y^2}{9} = u^2 + v^2.$$

So the ellipse in (x, y) -space is the image of the unit disk

$$u^2 + v^2 \leq 1$$

in (u, v) -space.

The Jacobian matrix is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix},$$

so

$$|\det J| = 6.$$

Therefore

$$\text{Area(ellipse)} = \iint_{\text{ellipse}} 1 \, dA = \iint_{u^2+v^2 \leq 1} 6 \, du \, dv = 6\pi.$$

Linear Scaling Principle

If a linear map stretches by factors $a_1, \dots, a_n$ in perpendicular directions, then it multiplies n -dimensional volume by

$$|a_1 \cdots a_n|.$$

The ellipse example is just a disk stretched by 2 horizontally and by 3 vertically.

3 Large Example: Volume of the n -Ball

3.1 Definitions

The n -dimensional ball of radius R is

$$B_R^n = \{x \in \mathbb{R}^n : \|x\| \leq R\}.$$

Its boundary is the $(n-1)$ -dimensional sphere

$$S_R^{n-1} = \{x \in \mathbb{R}^n : \|x\| = R\}.$$

Define

$$V_n(R) = \text{Vol}_n(B_R^n), \quad C_n = V_n(1).$$

Scaling First

Since B_R^n is the image of B_1^n under the dilation

$$x = Ru,$$

the Jacobian determinant is R^n . Hence

$$V_n(R) = C_n R^n.$$

The main problem is to compute the unit-ball constant C_n .

3.2 Low-Dimensional Starting Points

For $n = 1$,

$$B_R^1 = [-R, R], \quad V_1(R) = 2R.$$

Thus

$$C_1 = 2.$$

For $n = 2$,

$$B_R^2 = \{(x, y) : x^2 + y^2 \leq R^2\}, \quad V_2(R) = \pi R^2,$$

so

$$C_2 = \pi.$$

For $n = 3$,

$$V_3(R) = \frac{4}{3}\pi R^3,$$

so

$$C_3 = \frac{4}{3}\pi.$$

We now compute the general pattern.

3.3 Slicing Recursion

Write a point in $\mathbb{R}^n$ as

$$x = (x_1, \dots, x_{n-1}, t), \quad t = x_n.$$

At height t , the cross-section of B_R^n is

$$x_1^2 + \dots + x_{n-1}^2 \leq R^2 - t^2.$$

This is an $(n-1)$ -dimensional ball of radius

$$\sqrt{R^2 - t^2}.$$

Therefore

$$V_n(R) = \int_{-R}^R V_{n-1}(\sqrt{R^2 - t^2}) \, dt.$$

Using the scaling formula

$$V_{n-1}(s) = C_{n-1} s^{n-1},$$

we get

$$V_n(R) = C_{n-1} \int_{-R}^R (R^2 - t^2)^{(n-1)/2} \, dt.$$

3.4 Reducing to a Trigonometric Integral

Set

$$t = R \sin \theta.$$

Then

$$dt = R \cos \theta \, d\theta, \quad R^2 - t^2 = R^2 \cos^2 \theta.$$

As t goes from $-R$ to R , θ goes from $-\pi/2$ to $\pi/2$. Thus

$$\begin{aligned} V_n(R) &= C_{n-1} \int_{-\pi/2}^{\pi/2} (R^2 \cos^2 \theta)^{(n-1)/2} R \cos \theta \, d\theta \\ &= C_{n-1} R^n \int_{-\pi/2}^{\pi/2} \cos^n \theta \, d\theta. \end{aligned}$$

Define

$$I_n = \int_{-\pi/2}^{\pi/2} \cos^n \theta \, d\theta.$$

Then

$$V_n(R) = C_{n-1} I_n R^n.$$

Since $V_n(R) = C_n R^n$, we have the recursion

$$\boxed{C_n = C_{n-1} I_n.}$$

3.5 Computing I_n

The first two values are

$$I_0 = \int_{-\pi/2}^{\pi/2} 1 \, d\theta = \pi, \quad I_1 = \int_{-\pi/2}^{\pi/2} \cos \theta \, d\theta = 2.$$

For $n \geq 2$, integrate by parts:

$$I_n = \int_{-\pi/2}^{\pi/2} \cos^{n-1} \theta \cos \theta \, d\theta.$$

Take

$$u = \cos^{n-1} \theta, \quad dv = \cos \theta \, d\theta.$$

Then

$$du = -(n-1) \cos^{n-2} \theta \sin \theta \, d\theta, \quad v = \sin \theta.$$

The boundary term is

$$\cos^{n-1} \theta \sin \theta \Big|_{-\pi/2}^{\pi/2} = 0,$$

because $\cos(\pm\pi/2) = 0$ when $n \geq 2$. Hence

$$\begin{aligned} I_n &= (n-1) \int_{-\pi/2}^{\pi/2} \sin^2 \theta \cos^{n-2} \theta \, d\theta \\ &= (n-1) \int_{-\pi/2}^{\pi/2} (1 - \cos^2 \theta) \cos^{n-2} \theta \, d\theta \\ &= (n-1)(I_{n-2} - I_n). \end{aligned}$$

Therefore

$$nI_n = (n-1)I_{n-2},$$

or

$$\boxed{I_n = \frac{n-1}{n} I_{n-2}.}$$

What the Recursion Gives

For even powers:

$$I_{2m} = \frac{2m-1}{2m} \frac{2m-3}{2m-2} \cdots \frac{1}{2} \pi.$$

For odd powers:

$$I_{2m+1} = \frac{2m}{2m+1} \frac{2m-2}{2m-1} \cdots \frac{2}{3} \cdot 2.$$

This already computes every C_n recursively.

3.6 Closed Formula with the Gamma Function

The Gamma function satisfies

$$\Gamma(s+1) = s\Gamma(s), \quad \Gamma(1) = 1, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Thus $\Gamma(k+1) = k!$ for integers $k \geq 0$.

A standard form of the trigonometric integral is

$$I_n = \int_{-\pi/2}^{\pi/2} \cos^n \theta \, d\theta = \frac{\sqrt{\pi} \Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2} + 1\right)}.$$

Now use induction. Suppose

$$C_{n-1} = \frac{\pi^{(n-1)/2}}{\Gamma\left(\frac{n-1}{2} + 1\right)} = \frac{\pi^{(n-1)/2}}{\Gamma\left(\frac{n+1}{2}\right)}.$$

Then

$$\begin{aligned} C_n &= C_{n-1} I_n \\ &= \frac{\pi^{(n-1)/2}}{\Gamma\left(\frac{n+1}{2}\right)} \cdot \frac{\sqrt{\pi} \Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2} + 1\right)} \\ &= \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2} + 1\right)}. \end{aligned}$$

Therefore

$$V_n(R) = \text{Vol}_n(B_R^n) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2} + 1\right)} R^n.$$

Special Cases

$$V_1(R) = 2R, \quad V_2(R) = \pi R^2, \quad V_3(R) = \frac{4}{3} \pi R^3.$$

$$V_4(R) = \frac{\pi^2}{2} R^4, \quad V_5(R) = \frac{8\pi^2}{15} R^5.$$

4 Surface Area of the High-Dimensional Sphere

4.1 Shell Argument

Let

$$A_{n-1}(R) = \text{Area}(S_R^{n-1})$$

be the $(n-1)$ -dimensional surface area of the boundary of B_R^n .

A thin shell of radius R and thickness ΔR has approximate volume

$$A_{n-1}(R)\Delta R.$$

On the other hand, its volume is approximately

$$V_n(R + \Delta R) - V_n(R).$$

Dividing by ΔR and letting $\Delta R \rightarrow 0$ gives

$$A_{n-1}(R) = \frac{d}{dR} V_n(R).$$

4.2 Formula

Since

$$V_n(R) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} R^n,$$

we get

$$A_{n-1}(R) = \frac{n\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} R^{n-1}.$$

Using

$$\Gamma\left(\frac{n}{2} + 1\right) = \frac{n}{2} \Gamma\left(\frac{n}{2}\right),$$

this becomes

$$A_{n-1}(R) = \frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})} R^{n-1}.$$

Checks

$$A_0(R) = 2$$

counts the two endpoints of the interval $[-R, R]$.

$$A_1(R) = 2\pi R$$

is the circumference of a circle.

$$A_2(R) = 4\pi R^2$$

is the usual surface area of a sphere in $\mathbb{R}^3$.

$$A_3(R) = 2\pi^2 R^3$$

is the three-dimensional surface area of the sphere $S_R^3 \subset \mathbb{R}^4$.

Dimension Language

The ball B_R^n has dimension n . Its boundary S_R^{n-1} has dimension $n - 1$ but lives inside $\mathbb{R}^n$. Thus S_R^2 is the ordinary sphere in $\mathbb{R}^3$.

5 Average Values and Harmonic Functions

5.1 Average over a Region

For a finite region $D \subset \mathbb{R}^n$, the average value of f over D is

$$f_{\text{avg}, D} = \frac{1}{\text{Vol}_n(D)} \int_D f(x) \, dV_n.$$

For a sphere $S_R^{n-1}(a)$, the spherical average is

$$f_{\text{avg}, S} = \frac{1}{\text{Area}(S_R^{n-1}(a))} \int_{S_R^{n-1}(a)} f(x) \, dS.$$

5.2 Harmonic Functions

A twice differentiable function u on a region in $\mathbb{R}^n$ is harmonic if

$$\Delta u = \frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_n^2} = 0.$$

Theorem 5.1 (Mean Value Property). *If u is harmonic on a ball containing $S_R^{n-1}(a)$ and its interior, then*

$$\frac{1}{\text{Area}(S_R^{n-1}(a))} \int_{S_R^{n-1}(a)} u(x) \, dS = u(a).$$

In fact, the average of u over the full ball $B_R^n(a)$ also equals $u(a)$.

Meaning

For harmonic functions, the value at the center is not just a local value. It equals the average value around the center. This is why harmonic functions model equilibrium states.

5.3 Example 1: A Linear Harmonic Function

Let

$$u(x_1, \dots, x_n) = x_1 - 2x_2 + 7.$$

All second partial derivatives are zero, so

$$\Delta u = 0.$$

Therefore the average of u over any sphere centered at

$$a = (a_1, \dots, a_n)$$

is

$$u(a) = a_1 - 2a_2 + 7.$$

5.4 Example 2: A Quadratic Harmonic Function

In $\mathbb{R}^n$ with $n \geq 3$, consider

$$u(x_1, \dots, x_n) = x_1^2 - x_2^2 + 3x_3 + 5.$$

Then

$$\Delta u = 2 - 2 + 0 + \dots + 0 = 0,$$

so u is harmonic.

Let

$$a = (1, 2, 0, \dots, 0).$$

By the mean value property, the average of u over any sphere centered at a is

$$u(a) = 1^2 - 2^2 + 3 \cdot 0 + 5 = 2.$$

Thus

$$\boxed{\frac{1}{\text{Area}(S_R^{n-1}(a))} \int_{S_R^{n-1}(a)} (x_1^2 - x_2^2 + 3x_3 + 5) \, dS = 2.}$$

The same value is obtained for the average over the ball $B_R^n(a)$.

5.5 Same Example by Symmetry

Write a point on the sphere as

$$x = a + R\xi, \quad \|\xi\| = 1.$$

By symmetry on the unit sphere,

$$\text{avg}(\xi_i) = 0, \quad \text{avg}(\xi_i^2) = \frac{1}{n}.$$

Therefore

$$\text{avg}(x_i) = a_i,$$

and

$$\text{avg}(x_i^2) = a_i^2 + \frac{R^2}{n}.$$

Then

$$\begin{aligned} \text{avg}(x_1^2 - x_2^2 + 3x_3 + 5) &= \left(a_1^2 + \frac{R^2}{n}\right) - \left(a_2^2 + \frac{R^2}{n}\right) + 3a_3 + 5 \\ &= a_1^2 - a_2^2 + 3a_3 + 5 \\ &= 2. \end{aligned}$$

The R^2/n terms cancel exactly because the function is harmonic.

Contrast

The function $v(x) = \|x\|^2$ is not harmonic because

$$\Delta v = 2n.$$

On $S_R^{n-1}(0)$ its average is R^2 , not $v(0) = 0$. The mean value property is special to harmonic functions.

6 Physics Applications

6.1 Mass and Center of Mass

If $\rho(x)$ is density on a region $D \subset \mathbb{R}^n$, then the total mass is

$$M = \int_D \rho(x) \, dV_n.$$

The center of mass is

$$\bar{x} = \frac{1}{M} \int_D x \rho(x) \, dV_n.$$

In three dimensions, this means

$$\bar{x} = \frac{1}{M} \iiint_E x \rho \, dV, \quad \bar{y} = \frac{1}{M} \iiint_E y \rho \, dV, \quad \bar{z} = \frac{1}{M} \iiint_E z \rho \, dV.$$

Computation Pattern

1. Choose coordinates for the region.

2. Compute mass:

$$M = \int \rho \, dV.$$

3. Compute first moments:

$$\int x\rho \, dV, \quad \int y\rho \, dV, \quad \int z\rho \, dV.$$

4. Divide by M .

6.2 Torque

Torque is the rotational effect of a force. In Chinese, torque is usually called

力矩

when we mean “moment of a force.” It is also sometimes called

转矩

when the emphasis is on twisting or rotation.

If a force F is applied at position r relative to a chosen reference point, the torque is

$$\tau = r \times F.$$

The direction of τ is perpendicular to the plane of rotation, following the right-hand rule.

Torque and Balance

For a mass distribution in a uniform gravitational field g , the force on a small mass element is

$$dF = g \, dm.$$

The torque about a point R is

$$d\tau = (r - R) \times g \, dm.$$

The total torque is

$$\tau(R) = \int_D (r - R) \times g \, dm.$$

Since g is constant,

$$\tau(R) = \left(\int_D (r - R) \, dm \right) \times g.$$

Thus if

$$\int_D (r - R) \, dm = 0,$$

then

$$\tau(R) = 0.$$

This is the vector balance definition of the center of mass:

$$R = \frac{1}{M} \int_D r \, dm.$$

Small Caveat

If gravity points in one fixed direction, zero torque can hold for any point on the vertical line through the center of mass. The unique center of mass is obtained by requiring

$$\int_D (r - R) \, dm = 0,$$

or by requiring zero torque for every uniform force direction.

6.3 Moment of Inertia

The moment of inertia around the z -axis is

$$I_z = \iiint_E (x^2 + y^2) \rho(x, y, z) \, dV.$$

In cylindrical coordinates,

$$x^2 + y^2 = r^2, \quad dV = r \, dr \, d\theta \, dz.$$

Therefore

$$I_z = \iiint_{E^*} r^2 \rho(r \cos \theta, r \sin \theta, z) r \, dr \, d\theta \, dz.$$

The first r^2 is distance-squared from the axis. The extra r is the Jacobian.

7 Probability and the High-Dimensional Gaussian

7.1 Probability Density

A probability density p on $D \subset \mathbb{R}^n$ satisfies

$$\int_D p(x) \, dV_n = 1.$$

For an event region $A \subset D$,

$$P(A) = \int_A p(x) \, dV_n.$$

The average, or expected value, of a function f is

$$E[f] = \int_D f(x)p(x) \, dV_n.$$

7.2 The Standard Gaussian in $\mathbb{R}^n$

The standard Gaussian density in $\mathbb{R}^n$ is

$$p(x) = \frac{1}{(2\pi)^{n/2}} e^{-\|x\|^2/2}.$$

We now verify the normalization:

$$\int_{\mathbb{R}^n} e^{-\|x\|^2/2} \, dV_n = (2\pi)^{n/2}.$$

For any radial function $f(\|x\|)$, high-dimensional spherical coordinates give

$$\int_{\mathbb{R}^n} f(\|x\|) \, dV_n = A_{n-1}(1) \int_0^\infty f(r)r^{n-1} \, dr,$$

where

$$A_{n-1}(1) = \frac{2\pi^{n/2}}{\Gamma(n/2)}.$$

Apply this to $f(r) = e^{-r^2/2}$:

$$\int_{\mathbb{R}^n} e^{-\|x\|^2/2} \, dV_n = \frac{2\pi^{n/2}}{\Gamma(n/2)} \int_0^\infty e^{-r^2/2} r^{n-1} \, dr.$$

Let

$$s = \frac{r^2}{2}.$$

Then

$$r^{n-1} \, dr = 2^{n/2-1} s^{n/2-1} \, ds.$$

Therefore

$$\begin{aligned} \int_0^\infty e^{-r^2/2} r^{n-1} \, dr &= 2^{n/2-1} \int_0^\infty e^{-s} s^{n/2-1} \, ds \\ &= 2^{n/2-1} \Gamma(n/2). \end{aligned}$$

So

$$\int_{\mathbb{R}^n} e^{-\|x\|^2/2} \, dV_n = \frac{2\pi^{n/2}}{\Gamma(n/2)} \cdot 2^{n/2-1} \Gamma(n/2) = (2\pi)^{n/2}.$$

Normalized Density

Thus

$$p(x) = \frac{1}{(2\pi)^{n/2}} e^{-\|x\|^2/2}$$

has total integral 1 over $\mathbb{R}^n$.

7.3 Probability Inside a Ball

Let X have the standard Gaussian distribution in $\mathbb{R}^n$. Then

$$P(\|X\| \leq a) = \frac{1}{(2\pi)^{n/2}} \int_{\|x\| \leq a} e^{-\|x\|^2/2} dV_n.$$

Using radial coordinates,

$$P(\|X\| \leq a) = \frac{A_{n-1}(1)}{(2\pi)^{n/2}} \int_0^a e^{-r^2/2} r^{n-1} dr.$$

Substituting the formula for $A_{n-1}(1)$ gives

$$P(\|X\| \leq a) = \frac{1}{2^{n/2-1} \Gamma(n/2)} \int_0^a e^{-r^2/2} r^{n-1} dr.$$

Equivalently, with $s = r^2/2$,

$$P(\|X\| \leq a) = \frac{\gamma\left(\frac{n}{2}, \frac{a^2}{2}\right)}{\Gamma(n/2)},$$

where

$$\gamma(\alpha, b) = \int_0^b e^{-s} s^{\alpha-1} ds$$

is the lower incomplete Gamma function.

Special case: $n = 2$. For $n = 2$,

$$P(\|X\| \leq a) = \int_0^a e^{-r^2/2} r dr = 1 - e^{-a^2/2}.$$

7.4 High-Dimensional Geometry of the Gaussian

The radial density of $R = \|X\|$ is

$$q_n(r) = \frac{1}{2^{n/2-1} \Gamma(n/2)} r^{n-1} e^{-r^2/2}, \quad r \geq 0.$$

The exponential term wants r small, but the geometric factor r^{n-1} counts how much surface area exists at radius r .

To find the most likely radius, maximize

$$\log q_n(r) = \text{constant} + (n-1) \log r - \frac{r^2}{2}.$$

Differentiate:

$$\frac{d}{dr} \log q_n(r) = \frac{n-1}{r} - r.$$

Setting this equal to zero gives

$$r = \sqrt{n-1}.$$

Also,

$$E[\|X\|^2] = n.$$

So a high-dimensional standard Gaussian usually has radius on the scale

$$\sqrt{n}.$$

Probability Lesson

In high dimension, most Gaussian mass is not near the origin. It concentrates in a shell whose radius is approximately $\sqrt{n}$. This is a direct interaction between analysis and geometry: density decreases with radius, but surface area grows with radius.

8 Beginning Line Integrals

8.1 Curves and Arc Length

Let a curve C be parametrized by

$$r(t) = (x(t), y(t), z(t)), \quad a \leq t \leq b.$$

A small change Δt produces a small displacement

$$r'(t)\Delta t.$$

Thus the small arc length is

$$ds = \|r'(t)\| dt.$$

8.2 Scalar Line Integrals

If f is a scalar function on a curve C , then

$$\int_C f ds = \int_a^b f(r(t)) \|r'(t)\| dt.$$

If f is density, this gives the mass of a wire.

Example. Let C be the line segment from $(0, 0)$ to $(1, 1)$, and let

$$f(x, y) = x + y.$$

Parametrize the segment by

$$r(t) = (t, t), \quad 0 \leq t \leq 1.$$

Then

$$\|r'(t)\| = \|(1, 1)\| = \sqrt{2}.$$

So

$$\int_C (x + y) \, ds = \int_0^1 (2t)\sqrt{2} \, dt = \sqrt{2}.$$

8.3 Vector Line Integrals

A vector field assigns a vector to each point in space. For example, in three dimensions,

$$F(x, y, z) = (P(x, y, z), Q(x, y, z), R(x, y, z)).$$

At every point (x, y, z) , the vector field gives an arrow. Physically, this arrow may represent force, velocity of a fluid, electric field, wind direction, or any quantity with both magnitude and direction.

Scalar Function vs. Vector Field

A scalar function assigns a number to each point:

$$f(x, y, z) \in \mathbb{R}.$$

A vector field assigns a vector to each point:

$$F(x, y, z) \in \mathbb{R}^3.$$

So a scalar function can describe temperature or density, while a vector field can describe force or flow.

If F is a vector field, the vector line integral is

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) \, dt.$$

In physics, this computes work:

$$\text{work} = \int_C F \cdot dr.$$

The dot product measures the component of force tangent to the motion.

Two Line Integrals

$$\int_C f \, ds \quad \text{uses length and ignores orientation.}$$

$$\int_C F \cdot dr \quad \text{uses direction and changes sign if the path orientation is reversed.}$$

8.4 Geometry and Physics of Line Integrals

There are two common types of line integrals, and they answer different physical questions.

Type 1: Scalar Line Integral

$$\int_C f \, ds$$

adds up a scalar density along a curve. The small piece is length:

$$ds = \|r'(t)\| \, dt.$$

Geometrically, if $f = 1$, then

$$\int_C 1 \, ds = \text{length of } C.$$

If $f \geq 0$, we can imagine a “curtain” standing over the curve, with height f . The integral is the area of that curtain.

Physically, if f is mass density on a thin wire, then

$$\int_C f \, ds = \text{mass of the wire.}$$

This type does not care about orientation, because length is always positive.

Type 2: Vector Line Integral

$$\int_C F \cdot dr$$

adds up the component of a vector field in the direction of motion. Since

$$dr = r'(t) \, dt,$$

we have

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) \, dt.$$

Physically, if F is a force field, this is work. Only the tangential component of the force contributes. A force perpendicular to the motion does no work.

The connection between the two types comes from the unit tangent vector

$$T(t) = \frac{r'(t)}{\|r'(t)\|}.$$

Since

$$dr = T \, ds,$$

we get

$$\boxed{\int_C F \cdot dr = \int_C (F \cdot T) \, ds.}$$

So a vector line integral is a scalar line integral of the tangential component $F \cdot T$.

Connection

scalar line integral: $\int_C \text{density} \times \text{length}.$

vector line integral: $\int_C \text{tangential component} \times \text{length}.$

The first measures accumulated amount along a curve. The second measures signed effort along a direction. Reversing the curve leaves ds unchanged but changes T to $-T$, so the vector line integral changes sign.

8.5 Green's Theorem

Now specialize to a vector field in the plane:

$$F(x, y) = (P(x, y), Q(x, y)).$$

Along a plane curve C , the vector line integral becomes

$$\int_C F \cdot dr = \int_C P \, dx + Q \, dy.$$

If C is a closed curve, we often write

$$\oint_C P \, dx + Q \, dy.$$

This closed-path vector line integral is called the circulation; in Chinese, it is often called 环流 or 环量.

Positive Orientation

For a simple closed curve C bounding a region D , positive orientation means counterclockwise orientation. Equivalently, as you walk along C , the region D stays on your left.

Theorem 8.1 (Green's Theorem). *Let D be a plane region whose boundary $C = \partial D$ is a positively oriented, piecewise smooth, simple closed curve. Suppose P and Q have continuous first partial derivatives on a region containing D . Then*

$$\oint_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA.$$

Meaning

Green's theorem converts a circulation integral around the boundary into a double integral over the inside:

boundary circulation (边界环流) = total interior rotation.

The quantity

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

is the two-dimensional scalar curl of the vector field $F = (P, Q)$.

Divergence and Curl

For a plane vector field

$$F = (P, Q),$$

there are two important local quantities:

$$\operatorname{div} F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}, \quad \operatorname{curl} F = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}.$$

The divergence measures source or sink strength: how much the vector field flows outward from a point. The scalar curl measures rotation, also called vorticity: how much the vector field tends to spin around a point.

If T is the unit tangent and n is the outward unit normal along the positively oriented boundary ∂D , then Green's theorem has two useful forms:

$$\oint_{\partial D} F \cdot T \, ds = \iint_D \operatorname{curl} F \, dA \quad \text{circulation form,}$$

and

$$\oint_{\partial D} F \cdot n \, ds = \iint_D \operatorname{div} F \, dA \quad \text{flux form.}$$

The first says total boundary rotation equals total interior curl. The second says total outward flux equals total interior source.

Normal Direction Derivative

Let $u(x, y)$ be a scalar function, and let $C = \partial D$ be positively oriented. At a smooth point of the boundary, let

$$T = \left(\frac{dx}{ds}, \frac{dy}{ds} \right)$$

be the unit tangent vector. Since positive orientation keeps D on the left, the outward unit normal vector is

$$n = \left(\frac{dy}{ds}, -\frac{dx}{ds} \right).$$

The normal direction derivative is

$$\frac{\partial u}{\partial n} = \nabla u \cdot n.$$

It measures how fast u changes when we move perpendicular to the boundary, in the outward direction. Multiplying by ds gives

$$\frac{\partial u}{\partial n} ds = \nabla u \cdot n ds = u_x dy - u_y dx.$$

Therefore Green's theorem gives

$$\int_{\partial D} \frac{\partial u}{\partial n} ds = \oint_{\partial D} u_x dy - u_y dx = \iint_D (u_{xx} + u_{yy}) dA = \iint_D \Delta u dA.$$

So for a harmonic function, $\Delta u = 0$, the total outward normal derivative on the boundary is zero.

Green's Identities

Green's theorem also leads to identities used constantly in physics and PDE. Let u and v be smooth scalar functions. Applying the flux form to the vector field $u \nabla v$ gives Green's first identity:

$$\boxed{\iint_D \nabla u \cdot \nabla v dA + \iint_D u \Delta v dA = \int_{\partial D} u \frac{\partial v}{\partial n} ds.}$$

Exchanging u and v and subtracting gives Green's second identity:

$$\boxed{\iint_D (u \Delta v - v \Delta u) dA = \int_{\partial D} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds.}$$

These identities are integration by parts for two-dimensional regions. They are central in electrostatics, heat flow, fluid flow, harmonic functions, and boundary value problems for Laplace and Poisson equations.

Proof for simple regions. It is enough to prove the two identities

$$\oint_C P \, dx = - \iint_D \frac{\partial P}{\partial y} \, dA,$$

and

$$\oint_C Q \, dy = \iint_D \frac{\partial Q}{\partial x} \, dA.$$

Adding them gives Green's theorem.

First suppose D is Type I:

$$D = \{(x, y) : a \leq x \leq b, \, g_1(x) \leq y \leq g_2(x)\}.$$

The positively oriented boundary moves left to right along the lower curve $y = g_1(x)$ and right to left along the upper curve $y = g_2(x)$. The vertical side pieces contribute nothing to $\int_C P \, dx$ because $dx = 0$ there. Hence

$$\begin{aligned} \oint_C P \, dx &= \int_a^b P(x, g_1(x)) \, dx + \int_b^a P(x, g_2(x)) \, dx \\ &= \int_a^b (P(x, g_1(x)) - P(x, g_2(x))) \, dx. \end{aligned}$$

By the one-variable fundamental theorem of calculus in the y direction,

$$P(x, g_2(x)) - P(x, g_1(x)) = \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y}(x, y) \, dy.$$

Therefore

$$\oint_C P \, dx = - \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y}(x, y) \, dy \, dx = - \iint_D \frac{\partial P}{\partial y} \, dA.$$

Next suppose D is Type II:

$$D = \{(x, y) : c \leq y \leq d, \, h_1(y) \leq x \leq h_2(y)\}.$$

The positively oriented boundary moves upward along the right curve $x = h_2(y)$ and downward along the left curve $x = h_1(y)$. The horizontal side pieces contribute nothing to $\int_C Q \, dy$ because $dy = 0$ there. Thus

$$\begin{aligned} \oint_C Q \, dy &= \int_c^d Q(h_2(y), y) \, dy + \int_d^c Q(h_1(y), y) \, dy \\ &= \int_c^d (Q(h_2(y), y) - Q(h_1(y), y)) \, dy. \end{aligned}$$

By the fundamental theorem of calculus in the x direction,

$$Q(h_2(y), y) - Q(h_1(y), y) = \int_{h_1(y)}^{h_2(y)} \frac{\partial Q}{\partial x}(x, y) \, dx.$$

Therefore

$$\oint_C Q \, dy = \int_c^d \int_{h_1(y)}^{h_2(y)} \frac{\partial Q}{\partial x}(x, y) \, dx \, dy = \iint_D \frac{\partial Q}{\partial x} \, dA.$$

For a more general region, split D into finitely many simple pieces.

Why a finite split is allowed

This statement is not true for an arbitrary subset of the plane. Here we are working with the regular regions used in this course: bounded regions whose boundary consists of finitely many smooth arcs, with only finitely many corners and changes of direction. By drawing finitely many vertical or horizontal cut segments through those special boundary points, we remove the overhangs. Each remaining piece is a curvilinear rectangle or triangle that is simple in the coordinate directions, so the two identities above apply to each piece.

The double integrals over the pieces add. The line integrals over interior cut curves cancel because each interior curve is traversed twice in opposite directions. Only the outer boundary remains. This proves Green's theorem for the usual piecewise smooth regions used in this course. For more irregular domains, one needs a more careful version of Green's theorem, usually proved by approximation.

Example

Let C be the unit circle oriented counterclockwise. Compute

$$\oint_C -y \, dx + x \, dy.$$

Here

$$P(x, y) = -y, \quad Q(x, y) = x.$$

Thus

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - (-1) = 2.$$

By Green's theorem,

$$\oint_C -y \, dx + x \, dy = \iint_{x^2+y^2 \leq 1} 2 \, dA = 2\pi.$$

8.6 Application: Conservative Vector Fields

A vector field F is called conservative if it is the gradient of some scalar potential function:

$$F = \nabla \phi.$$

In Chinese, conservative vector field is 保守向量场. If

$$F(x, y) = (P(x, y), Q(x, y)),$$

then this means

$$P = \frac{\partial \phi}{\partial x}, \quad Q = \frac{\partial \phi}{\partial y}.$$

Path Independence

If $F = \nabla\phi$, then for any curve C from A to B ,

$$\int_C F \cdot dr = \int_C \nabla\phi \cdot dr = \phi(B) - \phi(A).$$

Therefore the integral depends only on the endpoints, not on the path. This is called path independence (路径无关).

Green's theorem gives a useful test in the plane. Suppose $F = (P, Q)$ is defined on a simply connected region, meaning the region has no holes. If

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0,$$

then Green's theorem gives, for every closed curve C bounding a region inside the domain,

$$\oint_C P \, dx + Q \, dy = \iint_D 0 \, dA = 0.$$

So the circulation around every closed curve is zero.

Now take two curves C_1 and C_2 from the same point A to the same point B . Follow C_1 from A to B , then follow C_2 backward from B to A . This makes a closed curve. Since the integral around this closed curve is zero,

$$0 = \int_{C_1} F \cdot dr - \int_{C_2} F \cdot dr.$$

Thus

$$\int_{C_1} F \cdot dr = \int_{C_2} F \cdot dr.$$

The line integral is independent of path.

Important Caveat

The phrase “simply connected” matters. On a domain with a hole, zero scalar curl does not always imply path independence. A standard example is

$$F(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$$

on $\mathbb{R}^2 \setminus \{(0, 0)\}$. Its scalar curl is zero away from the origin, but its circulation around the unit circle is 2π .

Example

Let

$$F(x, y) = (2xy, x^2 + 2y).$$

Then

$$\frac{\partial Q}{\partial x} = 2x, \quad \frac{\partial P}{\partial y} = 2x.$$

So the scalar curl is zero. Since the domain is all of $\mathbb{R}^2$, the vector field is conservative and line integrals are path independent.

In fact, a potential function is

$$\phi(x, y) = x^2y + y^2,$$

because

$$\nabla\phi = (2xy, x^2 + 2y) = F.$$

Therefore, for any curve from $(0, 0)$ to $(1, 1)$,

$$\int_C F \cdot dr = \phi(1, 1) - \phi(0, 0) = 2.$$

9 Beginning Surface Integrals

9.1 Parametrized Surfaces

Let a surface S be parametrized by

$$X(u, v) = (x(u, v), y(u, v), z(u, v)), \quad (u, v) \in D.$$

The two tangent vectors are

$$X_u = \frac{\partial X}{\partial u}, \quad X_v = \frac{\partial X}{\partial v}.$$

The small area element is

$$dS = \|X_u \times X_v\| \, du \, dv.$$

9.2 Scalar Surface Integrals

For a scalar function f on a surface,

$$\iint_S f \, dS = \iint_D f(X(u, v)) \|X_u \times X_v\| \, du \, dv.$$

If $f = 1$, this gives surface area.

Example: Graph Surface. If

$$S : z = g(x, y), \quad (x, y) \in D,$$

then a parametrization is

$$X(x, y) = (x, y, g(x, y)).$$

Here

$$X_x = (1, 0, g_x), \quad X_y = (0, 1, g_y).$$

Thus

$$X_x \times X_y = (-g_x, -g_y, 1),$$

and

$$dS = \sqrt{1 + g_x^2 + g_y^2} \, dx \, dy.$$

So

$$\text{Area}(S) = \iint_D \sqrt{1 + g_x^2 + g_y^2} \, dx \, dy.$$

9.3 Flux Surface Integrals

If F is a vector field, the flux through an oriented surface is

$$\iint_S F \cdot n \, dS.$$

Using a parametrization, the oriented area vector is

$$(X_u \times X_v) \, du \, dv.$$

Therefore

$$\iint_S F \cdot n \, dS = \iint_D F(X(u, v)) \cdot (X_u \times X_v) \, du \, dv.$$

Changing the orientation changes the sign of the flux.

Next Topic

Line integrals and surface integrals continue the same story:

$$\text{integral} = \int \text{density} \times \text{small geometric size}.$$

For curves, the small size is ds or dr . For surfaces, it is dS or the oriented vector area element.

10 Practice Problems

1. Compute the volume of

$$x^2 + y^2 \leq z \leq 9$$

using cylindrical coordinates.

2. Use the change of variables

$$x = au, \quad y = bv$$

to compute the area of

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1.$$

3. Derive $V_4(R)$ from the formula

$$V_n(R) = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)} R^n.$$

4. Compute the surface area of $S_R^3 \subset \mathbb{R}^4$.

5. Let

$$u(x, y, z) = x^2 - y^2 + z.$$

Find the average of u over the sphere of radius 5 centered at

$$(1, 2, 3).$$

6. For the standard Gaussian in $\mathbb{R}^2$, compute

$$P(\|X\| \leq 2).$$

7. Compute

$$\int_C (x + y) \, ds$$

where C is the line segment from $(1, 0)$ to $(0, 1)$.

8. Use Green's theorem to compute

$$\oint_C y \, dx + x^2 \, dy,$$

where C is the positively oriented boundary of the rectangle

$$0 \leq x \leq 2, \quad 0 \leq y \leq 1.$$

9. Show that

$$F(x, y) = (2xy, x^2 + 2y)$$

is conservative, and compute $\int_C F \cdot dr$ for any curve C from $(0, 0)$ to $(1, 1)$.

10. Let S be the graph $z = x^2 + y^2$ over the disk $x^2 + y^2 \leq 1$. Set up, but do not evaluate, the surface area integral.

11 Summary

- Polar, cylindrical, and spherical coordinates are special cases of the general change-of-variables formula.
- The Jacobian determinant is the local size-scaling factor.
- The n -dimensional ball has volume

$$\text{Vol}_n(B_R^n) = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)} R^n.$$

- The sphere S_R^{n-1} has surface area

$$\text{Area}(S_R^{n-1}) = \frac{2\pi^{n/2}}{\Gamma(n/2)} R^{n-1}.$$

- Harmonic functions satisfy the mean value property: their average over a sphere or ball equals the center value.
- Torque is 力矩 in Chinese; it measures the rotational effect of force.
- A high-dimensional Gaussian has most of its mass near radius $\sqrt{n}$.
- Scalar line integrals accumulate density along length; vector line integrals accumulate tangential components and model work or circulation.
- Green's theorem connects circulation with curl, and outward flux with divergence.
- Green's identities connect boundary normal derivatives with the Laplacian; they are basic tools in physics and PDE.
- On a simply connected region, zero scalar curl implies path independence for line integrals.
- Line integrals integrate along curves; surface integrals integrate over surfaces.